

GraphExplainer: Explaining nodes, edges and attributes of Graph Neural Network predictions

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Why Explain Graph Regression Models?

Graph Neural Networks (GNNs) are widely used for molecular property prediction, where understanding individual predictions is essential for scientific discovery and decision-making. Existing black-box explainers identify important nodes, edges, or subgraphs, but provide limited support for feature-level explanations and do not explain edge attributes. Moreover, current evaluation metrics are primarily designed for classification tasks and are not well suited to graph regression.

Goal: Develop a local black-box explainer for graph regression capable of interpreting nodes, edges, node attributes, and edge attributes, together with a fidelity metric specifically designed for regression explanations.

Key Limitation of Existing Explainers

Explanation Unit Existing Methods* GraphExplainer

Nodes	✓	✓
Node Attributes	Partial	✓
Edges	✓	✓
Edge Attributes	×	✓

Table: Research gap addressed by GraphExplainer. Existing Methods include GNNExplainer, SubgraphX, PGM-Explainer, RelEX, GraphSHAP and Dai.

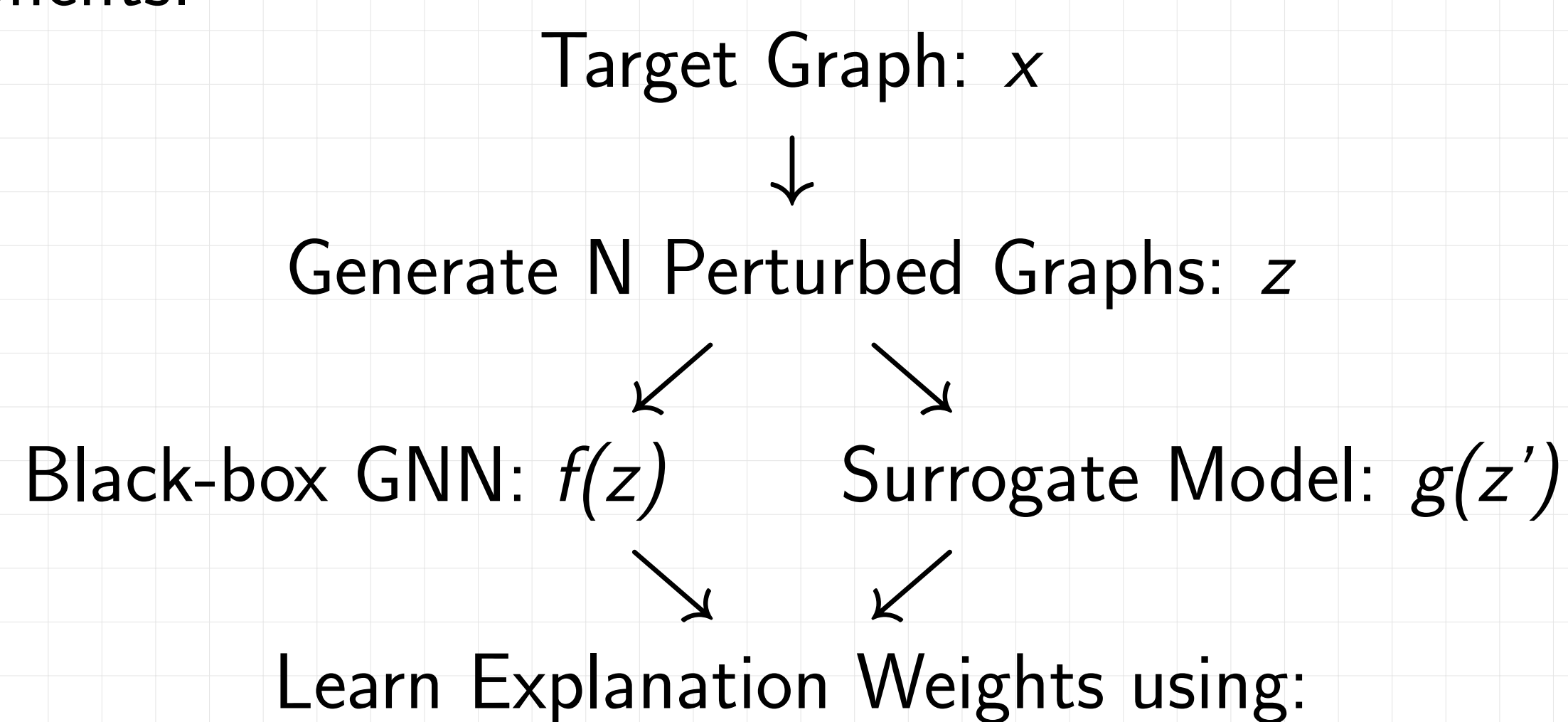
Edge attributes remain unexplained by existing black-box methods. GraphExplainer provides the first unified explanation of nodes, edges, node attributes and edge attributes.

Methods

1. Train AqSol dataset for solubility prediction using GINE
2. GraphExplainer pipeline
3. RegFidelity^k metric

GraphExplainer pipeline

GraphExplainer is a surrogate-based local explainer for graph regression. It approximates the behaviour of a black-box GNN around a target graph and attributes importance to graph components.



$$\alpha^*, \beta^*, \gamma^*, \delta^* = \underset{\alpha, \beta, \gamma, \delta}{\operatorname{argmin}} \left(\sum_{z, z' \in \mathcal{Z}} e^{-\frac{D(x, z)^2}{\sigma^2}} (f(z) - g(z'))^2 \right)$$

where α : nodes, β : node attributes, γ : edges and δ : edge attributes

RegFidelity_(u)^k metric

It measures explanation fidelity by progressively removing the k least important nodes, edges, node attributes, or edge attributes and evaluating the impact on the prediction. Higher AUC values indicate better explanations.

$$\operatorname{RegFidelity}_{(u)}^k = e^{-|f(G) - f(G_{(u)}^k)|}$$

Acknowledgements

This project was founded by the project PID2022-138327OB-I00 financed by the Ministerio de Ciencia e Innovación (MCIN)/Agencia Estatal de Investigación (AEI)/10.13039/501100011033/FEDER, UE.

Results: Comparison with existing explainers

GraphExplainer is compared with GNNExplainer using qualitative molecular explanations and the proposed RegFidelity^k metric.

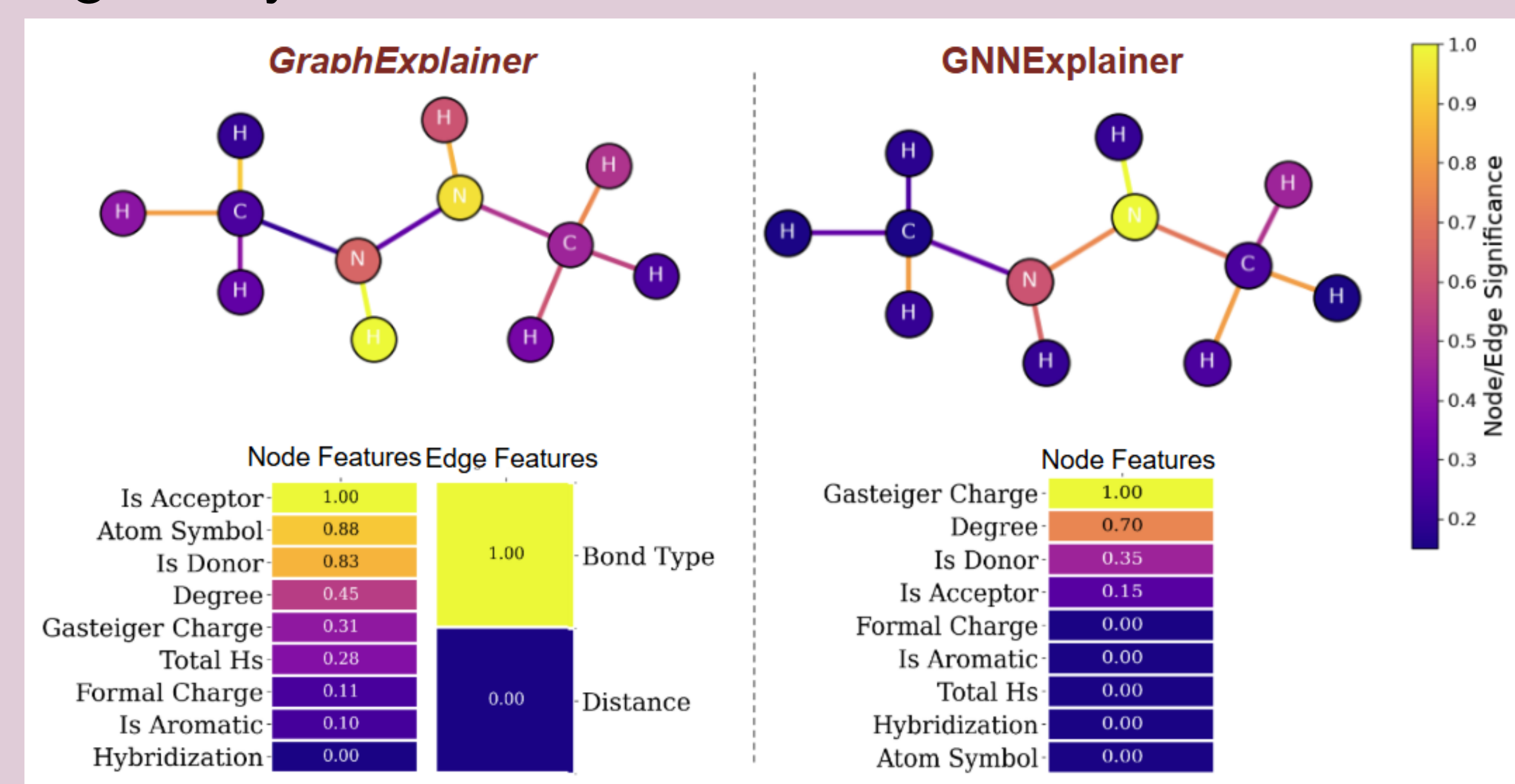


Figure: Interpretations of compound B-1122.

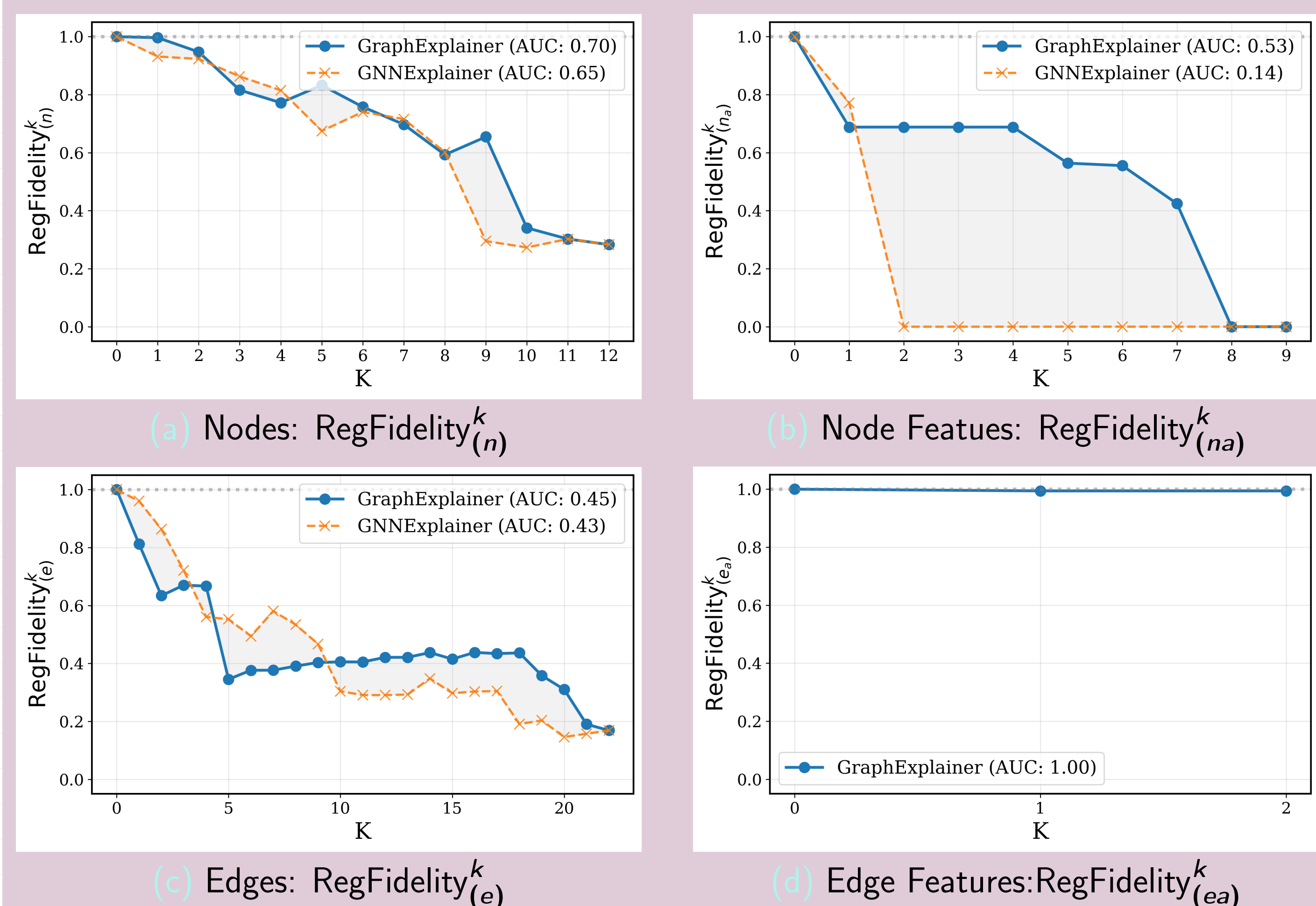


Figure: RegFidelity comparison over k on compound B-1122.

Method	Nodes	Node Atts	Edges	Edge Atts
GraphExplainer	0.40	0.53	0.52	0.79
GNNExplainer	0.44	0.13	0.55	-

Table: Mean of AUC of RegFidelity^k on test sets.

Only GraphExplainer additionally explains edge attributes and achieves higher fidelity for feature-level explanations.

Main Contributions

- First local black-box explainer for graph regression.
- Explains nodes, edges, node attributes, and edge attributes.
- RegFidelity^k: New regression-specific evaluation metric.
- Validated on molecular solubility prediction.